stress testing software is running. Using LN2 for cooling isn't easy for several reasons. You need to setup the motherboard with additional components for insulation to prevent any leakage or moisture build-up that could be fatal to the board. A specialised CPU cooling pot made of aluminium or copper needs to be installed that can contain LN2 while the lower part of the pot acts as the CPU block. The LN2 poured in the pot will be making direct contact with the CPU block on the other side inside and keeping the CPU temperature low. You'll need to ensure that there's enough insulation around the CPU pot and below the motherboard as well since condensation will continuously keep occuring throughout the process.

Once you've taken care of the motherboard, you need to prepare how you're going to store and pour LN2. Since you can't store it at room temperature, you'll be procuring LN2 in pressurised canisters. To make it easier to pour inside the pot, you will be using a regular thermos to temporarily store and pour the liquid. Attaching a thermometer to the CPU pot is essential since you need to monitor the temperature continuously. With everything setup, you can begin your LN2 cooling journey. You could also keep a blow torch handy in case you end up cooling the CPU too much. Otherwise, you'll have to wait until the temperature naturally starts to rise again. LN2 is used not just on CPUs but on the GPUs as well in their appropriate pots. Most of the process is a DIY method but EVGA recently showcased a closed loop or AIO (all-in-one) cooling system using LN2 as the coolant called the "Roboclocker". This makes it the world's first AIO liquid cooler to use liquid nitrogen. Of course, you'll still need to store LN2 canisters beside your rig.

Apart from that, liquid Helium or LHe is also finding its way into overclocking because of its extremely low boiling point of -269 degrees Celsius. But it's expensive and hardly seen at overclocking events. For beginners, dry ice could be a safe and inexpensive starting point. The same process is followed while the dry ice is crushed and mixed with acetone. This creates a uniform cooling substance that can be poured into the pot and the CPU can be overclocked.

Phase-change cooling

Liquid nitrogen or liquid helium is usually used by overclockers to push their CPU clocks to insane speeds. Another method that can be employed to achieve sub-zero temperatures for you CPU is by using phase-change cooling. Also called vapour cooling, it works on the same principle used in refrigerators and air-conditioners. In this type of cooling, a coolant or gas